

GANGARAPU BHAVANA

Java Backend Developer | Java | Spring Boot | REST APIs | PostgreSQL | Kafka | Redis
+91 7036782591 | gangarapubhavana945@gmail.com | LinkedIn | GitHub | Bengaluru, India

PROFESSIONAL SUMMARY

Java Backend Developer with 4 years of experience building and enhancing production applications using Java, Spring Boot, Spring MVC, REST APIs, Hibernate/JPA, and PostgreSQL. At General Motors, contributed to designing and developing a backend application from scratch and personally improved database query execution time from approximately 4 minutes to 0.9 seconds and API response time from approximately 5 minutes to 1 minute through query and data-access optimization. Experienced in OAuth2-based Single Sign-On, Jenkins, CI/CD, Python automation, production releases, Git/GitHub, and Agile development. Hands-on experience with Apache Kafka and Redis through backend projects.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate/JPA, REST APIs, OAuth2

Databases: PostgreSQL, MySQL

Messaging & Caching: Apache Kafka, Redis

DevOps & Tools: Jenkins, Git, GitHub, CI/CD, Azure DevOps, Postman, Insomnia, Eclipse, VS Code

Development Practices: OOP, Design Patterns, SOLID Principles, Data Structures & Algorithms, REST API Design, Agile/Scrum

Automation: Jira Automation, Python Automation

PROFESSIONAL EXPERIENCE

Systems Engineer | Tata Consultancy Services (TCS)

Client: General Motors | Bengaluru, India | Aug 2022 – Present

- Designed and developed backend features for a General Motors automotive diagnostics web application using Java, Spring Boot, Spring MVC, REST APIs, Hibernate/JPA, and PostgreSQL.
- Contributed to designing and developing the application from scratch, taking ownership of individual features from requirement analysis and implementation through testing and production release.
- Personally optimized PostgreSQL queries and Hibernate/JPA data-access logic, reducing database query execution time from approximately 4 minutes to 0.9 seconds.
- Optimized backend API processing and data-access flows, reducing API response time from approximately 5 minutes to 1 minute.
- Developed and maintained REST APIs for processing automotive diagnostic data and integrated OAuth2-based Single Sign-On (SSO) using General Motors Microsoft accounts.
- Participated in production releases and maintained Git/GitHub repositories for release-related development and maintenance.
- Developed and maintained Jenkins CI/CD automation and Python-based scripts supporting build, test, execution, and reporting workflows for General Motors hardware automation projects.
- Worked on Jira automation and administration activities, investigating assigned requirements and implementing workflow and release automation using Python.
- Collaborated with cross-functional teams in Agile/Scrum environments and contributed to successful production deliveries and client-appreciated work.

PROJECTS

URL Shortener System | Java, Spring Boot, Spring Security, PostgreSQL, Redis, Kafka, Docker

GitHub: github.com/Bhavana-source/url-shortener-system-springboot-redis-kafka

- Designed and developed a backend URL shortener using Spring Boot, REST APIs, PostgreSQL, Redis, Kafka, and Spring Security.
- Implemented URL creation, redirection, custom aliases, URL expiry, deactivation, and analytics functionality.
- Implemented Redis cache-aside caching to reduce repeated database reads and improve URL lookup performance.
- Integrated Apache Kafka for asynchronous click-tracking and analytics events, decoupling analytics processing from the core URL redirection flow.
- Implemented authentication and authorization using Spring Security and JWT-based authentication.
- Containerized the application using Docker Compose for local development and testing.
- Conducted application performance testing to evaluate caching, concurrency handling, and response-time improvements under simulated request loads.

CERTIFICATIONS

- Oracle Java Foundations – Learning Explorer | Oracle | Jun 2026
- Apache Kafka Fundamentals Accreditation | Confluent | 2026

EDUCATION

B.Tech, Electronics and Communication Engineering – Koneru Lakshmaiah University | CGPA: 8.6 / 10 | 2022